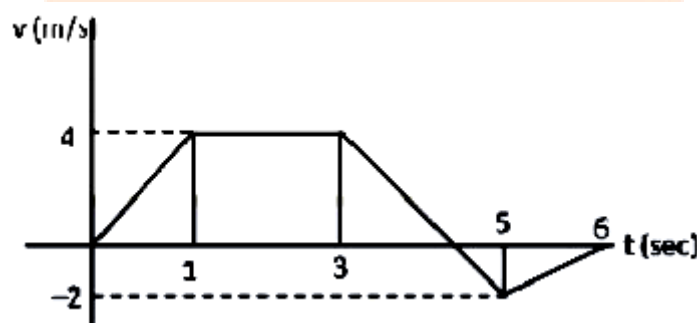


JEE MAIN 2020 5th SEPT SHIFT 2

Physics

Question 1. Velocity time graph of a particle is shown in figure. Find displacement of the particle.



- (1) 7
- (2) 11
- (3) 5
- (4) 15

Ans. (2)

Sol. $s = \text{area of graph}$

$$= \frac{1}{2} \times 4 \left(2 + \frac{13}{3} \right) - \frac{1}{2} \times 2 \times \frac{5}{3} = \frac{38}{3} - \frac{5}{3} = 11 \text{ m}$$

Question 2. A light of 4 eV incident on metal surface of work function ϕ_i eV. Another light of 2.5 eV incident on

another metal surface of work function ϕ_2 eV. Find the ratio of maximum velocity of photo electrons $\left(\frac{v_1}{v_2} \right)$.

(1) $\sqrt{\frac{4 + \phi_1}{2.5 - \phi_2}}$

(2) $\sqrt{\frac{4 - \phi_1}{2.5 - \phi_2}}$

(3) $\sqrt{\frac{3 - \phi_1}{2.5 - \phi_2}}$

(4) $\sqrt{\frac{6 + \phi_1}{2.5 - \phi_2}}$

Ans. (2)

Sol. $\frac{1}{2}mv^2 = E - \phi$

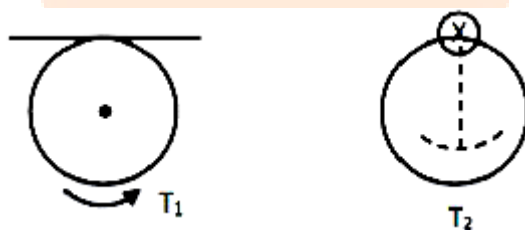
For 1st metal $\frac{1}{2}mv_1^2 = 4 - \phi_1$ (1)

For 2nd metal $\frac{1}{2}mv_2^2 = 2.5 - \phi_2$ (2)

$$\left(\frac{v_1}{v_2} \right)^2 = \frac{4 - \phi_1}{2.5 - \phi_2}$$

$$\frac{v_1}{v_2} = \sqrt{\frac{4 - \phi_1}{2.5 - \phi_2}}$$

Question 3. A ring-1 oscillate with period T_1 about tangential axis in the plane of ring and an another ring-2 oscillate with period T_2 about tangential axis perpendicular to plane of ring. $\frac{T_1}{T_2} = ?$



(1) $\sqrt{\frac{3}{4}}$

(2) $\sqrt{\frac{5}{8}}$

(3) $\sqrt{\frac{8}{5}}$

(4) $\sqrt{\frac{4}{3}}$

Ans. (1)

$$\text{Sol. } \frac{T_1}{T_2} = \sqrt{\frac{I_1}{I_2}} = \sqrt{\frac{\frac{3}{2}MR^2}{2MR^2}} = \sqrt{\frac{3}{4}}$$

Question 4. A car is moving towards a fixed wall. It blows horn of frequency of 440 Hz. The frequency of reflected sound observed by the driver is 480 Hz then find the speed of car in km/hr. (Speed of sound is 350 m/sec):

- (1) 64.78 km/hr
- (2) 26.78 km/hr
- (3) 54.78 km/hr
- (4) 47.78 km/hr

Ans. (3)

$$\text{Sol. } f_f'' = \left(\frac{V + V_c}{V - V_c} \right) f$$

$$480 = \left[\frac{350 - V_c}{350 - V_c} \right] 400$$

$$\frac{12}{11} = \frac{350 + V_c}{350 - V_c}$$

$$350 \times 12 - 12V_c = 350 \times 11 + 11V_c$$

$$350(12 - 11) = 23V_c$$

$$350 = 23V_c$$

$$V_c = \frac{350}{23} \text{ m/s}$$

$$V_c = \frac{350}{23} \times \frac{18}{5}$$

$$VC = 54.78 \text{ km/hr}$$

Question 5. Given that $x = \frac{1}{\sqrt{\epsilon_0 \mu_0}}$, $y = \frac{E}{B}$ & $z = \frac{1}{RC}$. Which of the following is correct.

- (1) dimension of x and z will be same
- (2) dimension of y and z will be same
- (3) dimension of x and y will be same
- (4) dimension of x, y and z is different

Ans. (3)

Question 6. There are two rods of length ℓ_1 and ℓ_2 and coefficient of linear expansions are α_1 and α_2 respectively. Find equivalent coefficient of thermal expansion for their combination in series.

$$(1) \frac{\alpha_1 + \alpha_2}{2}$$

$$(2) \frac{\alpha_1 \ell_1 + \alpha_2 \ell_2}{\alpha_1 + \alpha_2}$$

$$(3) \frac{\alpha_1 \ell_1 + \alpha_2 \ell_2}{\ell_1 + \ell_2}$$

$$(4) \sqrt{\alpha_1 \alpha_2}$$

Ans. (3)

Sol. At $t^\circ\text{C}$; $\ell_{\text{eq}} = \ell_1 + \ell_2$

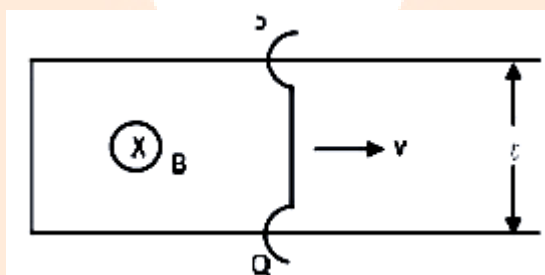
At $t + \Delta t^\circ\text{C}$; $\ell'_{\text{eq}} = \ell'_1 + \ell'_2$

$$\ell_{\text{eq}} (1 + \alpha_{\text{eq}} \Delta t) = \ell_1 (1 + \alpha_1 \Delta t) + \ell_2 (1 + \alpha_2 \Delta t)$$

$$(\ell_1 + \ell_2) (1 + \alpha_{\text{eq}} \Delta t) = \ell_1 + \ell_2 + \ell_1 \alpha_1 \Delta t + \ell_2 \alpha_2 \Delta t$$

$$\therefore \alpha_{\text{eq}} = \frac{\alpha_1 \ell_1 + \alpha_2 \ell_2}{\ell_1 + \ell_2}$$

Question 7. A rod having length l and resistance R is moving with velocity v on a π shape conductor. Find the current in the rod.



$$(1) \frac{1}{2} \frac{Bvl}{R}$$

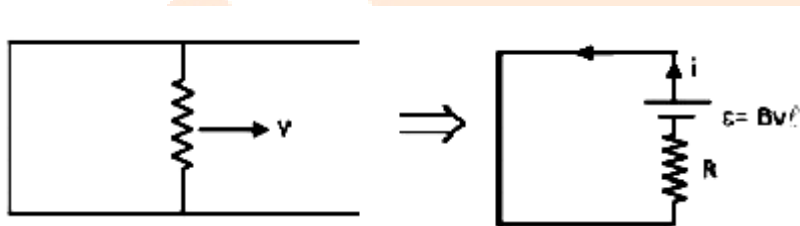
$$(2) \frac{2Bvl}{R}$$

$$(3) \frac{3Bvl}{R}$$

$$(4) \frac{Bvl}{R}$$

Ans. (4)

Sol.



$$i = \frac{Bvl}{R}$$

Question 8. There are two bodies A and B of same mass. A is placed near equator of earth and B is placed at a height 'h' above the pole of earth. If both the bodies weights equally. Find 'h' in terms of radius 'R' of earth, angular speed ' ω ' of earth and 'g' acceleration due to gravity close to earth.

$$(1) \frac{R\omega^2}{2g}$$

$$(2) \frac{R^2\omega^2}{2g}$$

$$(3) \frac{gR}{\omega^2}$$

$$(4) \frac{g}{\omega^2}$$

Ans. (2)

Sol. Both weight equally, if means effective 'g' is same for both

$$\text{For A } g_A = g - R\omega^2$$

$$\text{For B } g_B = g \left(1 - \frac{2h}{R} \right)$$

$$g_A = g_B$$

$$R\omega^2 = \frac{2g}{R}$$

$$\therefore h = \frac{R^2 \omega^2}{2g}$$

Question 9. Radioactive nucleus A decays into B with half-life 10 sec. A also convert into C with half-life 100 sec. Find half-life of A for both emission.

(1) 6 sec

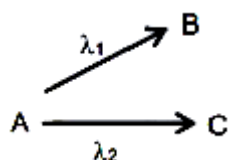
(2) 9 sec

(3) 3 sec

(4) 2 sec

Ans. (2)

Sol.



$$-\frac{dN}{dt} = \lambda_1 N + \lambda_2 N$$

$$-\frac{dN}{dt} = (\lambda_1 + \lambda_2) N$$

$$\lambda_{eq} = (\lambda_1 + \lambda_2)$$

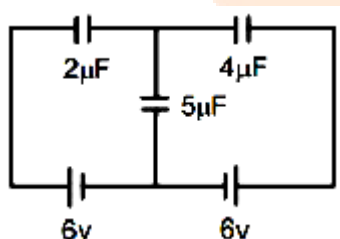
$$\frac{\ell \ln 2}{T} = \frac{\ell \ln 2}{T_1} + \frac{\ell \ln 2}{T_2}$$

$$\frac{1}{T} = \frac{1}{T_1} + \frac{1}{T_2}$$

$$\frac{1}{T} = \frac{1}{10} + \frac{1}{100} = \frac{11}{100}$$

$$T = \frac{100}{11} \approx 9 \text{ sec}$$

Question 10. Find charge on $5\mu\text{F}$.



(1) $\frac{120}{11} \mu\text{C}$

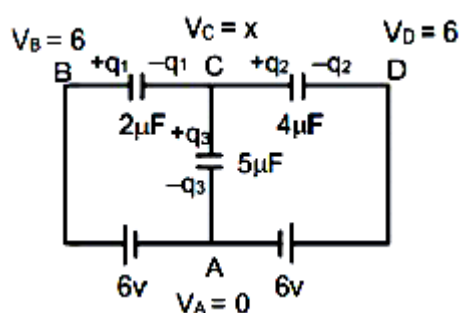
(2) $\frac{150}{11} \mu\text{C}$

(3) $\frac{180}{11} \mu\text{C}$

(4) $\frac{90}{11} \mu\text{C}$

Ans. (3)

Sol.



$$-q_1 + q_2 + q_3 = 0$$

$$-2(6 - x) + 4(x - 6) + 5(x - 0) = 0$$

$$-12 + 2x + 4x - 24 + 5x = 0$$

$$11x = 36 \Rightarrow x = \frac{36}{11}$$

$$q_3 = 5 \times \frac{36}{11} = \frac{180}{11} \mu\text{C}$$

Question 11. An ideal diatomic gas is taken through an adiabatic process in which density increases to 32 times. If pressure increases to 'n' times. Find n.

- (1) 4
- (2) 8
- (3) 64
- (4) 128

Ans. (4)

Sol. In adiabatic process

$$Pv^\gamma = \text{constant}$$

$$\therefore P \left(\frac{m}{\rho} \right)^\gamma = \text{constant}$$

As mass is constant

$$\therefore p \propto \rho^\gamma$$

$$\frac{P_f}{P_i} = \left(\frac{\rho_f}{\rho_i} \right)^\gamma = (32)^{7/5}$$

$$\therefore n = 2^7 = 128$$

Question 12. A body of mass 2 kg at rest is supplied constant power 1 J/sec., the distance travelled by the body after 6 sec. is:

- (1) $2\sqrt{6}m$
- (2) $4\sqrt{6}m$
- (3) $2\sqrt{3}m$

$$(4) 6\sqrt{3}m$$

Ans. (2)

Sol. $P = 1 \text{ J/sec}$

$$Pt = W = \Delta K$$

$$t = 1/2 m. v^2 = v^2$$

$$\therefore v = \sqrt{t} = \frac{ds}{dt}$$

$$\int_0^5 ds = \int_0^6 \sqrt{t} dt$$

$$S = \frac{2}{3} [t^{3/2}]_0^6$$

$$= \frac{2}{3} \times 6\sqrt{6} = 4\sqrt{6}m$$

Question 13. A dielectric having dielectric constant $K = 4$ is filled in a capacitor having plate length l and width b . Now length of capacitor is increased by l_1 for which energy stored becomes half of initial value. l_1 should be:

$$(1) 2l$$

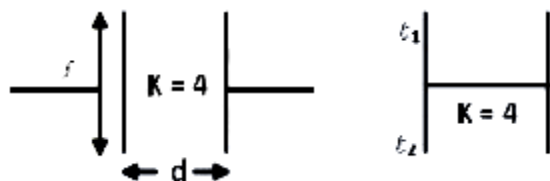
$$(2) 6l$$

$$(3) 8l$$

$$(4) 4l$$

Ans. (4)

Sol. Capacitance



$$\frac{Q^2}{2C_i} = 2 \frac{Q^2}{2C_f}$$

$$\frac{1}{C_i} = \frac{2}{C_f}$$

$$C_f = 2C_i$$

$$\frac{\epsilon_0}{d}(bl_1 + bl_4) = \frac{\epsilon_0}{d}bl_4 \times 2$$

$$l_1 + 41 = 81$$

$$l_1 = 41$$

Question 14. There is prism of refractive index 1.5 and prism angle 2° . Find minimum deviation caused by this prism.

- (1) 1°
- (2) 2°
- (3) $1/3^\circ$
- (4) $1/2^\circ$

Ans. (1)

Sol. For this prism gives minimum angle of deviation

$$\delta = (\mu - 1)A$$

$$\delta = (1.5 - 1) \times 2^\circ$$

$$= 1^\circ$$

Question 15. There is an iron core solenoid of turn density 10 turns/cm and volume 10^{-3}m^3 . It carries a current of 0.5 A and relative permeability of iron core is $\mu_r = 1000$. The magnetic moment of this solenoid is approximately (in A-m²)

(1) 5×10^2

(2) 5×10^3

(3) 5×10^4

(4) 5×10^5

Ans. (1)

Sol. Magnetic moment of an iron core solenoid

$$M = (\mu_r - 1).NiA$$

$$= (\mu_r - 1).Ni \frac{V}{l}$$

$$= (\mu - 1) \frac{N}{l} iV$$

$$= 999 \times \frac{10}{10^2} \times 0.5 \times 10^{-3}$$

$$= 499.5$$

Question 16. A ball is dropped from a height h . If falls on the liquid surface. Its velocity does not change when it enters in the liquid, find height h in terms of r = radius of ball, σ = density of liquid, ρ = density of ball, η = coefficient of viscosity and g = acceleration due to gravity:

$$(1) \frac{2}{81} \frac{r^4 g (\rho - \sigma)^2}{\eta^2}$$

$$(2) \frac{2}{50} \frac{r^4 g (\rho - \sigma)^2}{\eta^2}$$

$$(3) \frac{2}{25} \frac{r^4 g (\rho - \sigma)^2}{\eta^2}$$

$$(4) \frac{2}{90} \frac{r^4 g (\rho - \sigma)^2}{\eta^2} s$$

Ans. (1)

Sol. After falling through h the velocity should be equal to terminal velocity

$$\begin{aligned} \sqrt{2gh} &= \frac{2}{9} \frac{r^2 (\rho - \sigma) g}{\eta} \\ &= \frac{2}{81} \frac{r^4 g (\rho - \sigma)^2}{\eta^2} \end{aligned}$$

Question 17. A rocket moving in free space has varying mass due to fuel exhausted

$$\frac{dm(t)}{dt} = -bv^2(t)$$

where $m(t)$ = instantaneous mass

b = constant

$v(t)$ = instantaneous velocity

If gases are ejected with velocity u , with respect to rocket, instantaneous acceleration of rocket should be

(1) $\frac{ubv^2(t)}{m(t)}$

(2) $\frac{ubv^2(t)}{2m(t)}$

(3) $\frac{ubv(t)}{m(t)}$

(4) $\frac{ub}{m(t)}$

Ans. (1)

Sol. $F_{th} = u \frac{dm}{dt}$

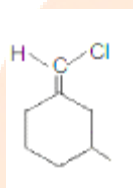
$$ma = u.bv^2$$

$$a = \frac{ubv^2}{m}$$

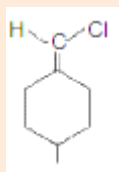
Chemistry

Question 18. Which of the following compound will show geometrical isomerism?

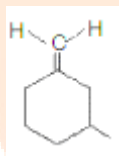
(1)



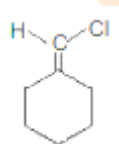
(2)



(3)



(4)



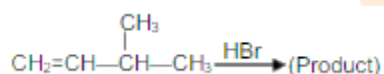
Answer: (1)

Solution:

Geometrical isomerism arises due to

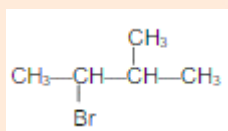
- (1) The presence of a restricted rotation (double bond or a ring structure).
- (2) Two different groups should be attached to any two carbon atoms of restricted rotation.

Question 19.

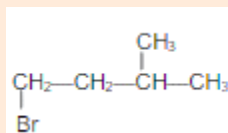


The product will be

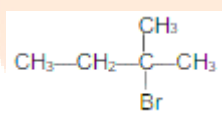
(1)



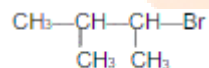
(2)



(3)

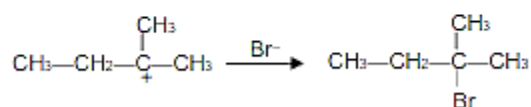
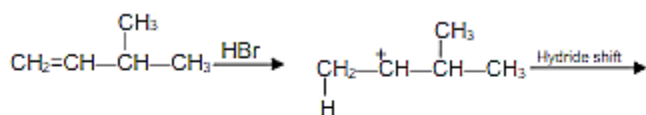


(4)



Answer: (3)

Solution:

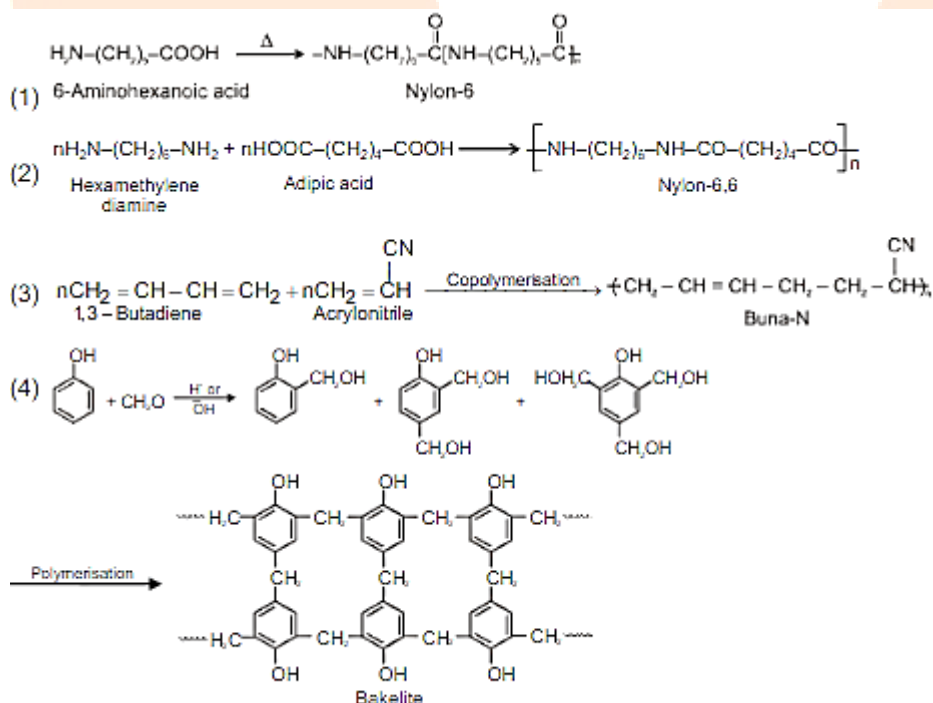


Question 20. Which is not a condensation polymer?

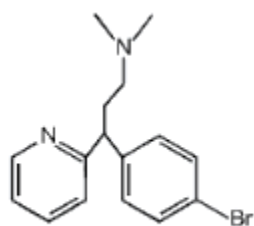
- (1) Nylon-6
- (2) Nylon-6, 6
- (3) Buna-N
- (4) Bakelite

Answer: (3)

Solution:



Question 21. What is the use of Brompheniramine?



Brompheniramine

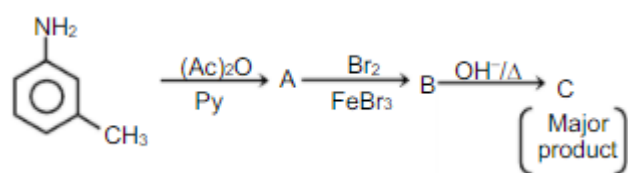
- (1) Antidepressant
- (2) Antihistamines
- (3) Antiseptic
- (4) Analgesic

Answer: (2)

Solution:

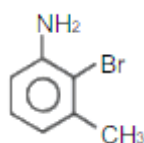
Synthetic drugs, brompheniramine (Dimetapp) act as antihistamines.

Question 22.

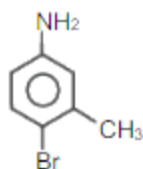


Product 'C' will be

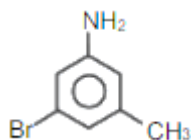
- (1)



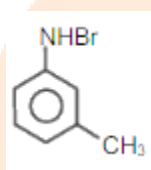
- (2)



(3)

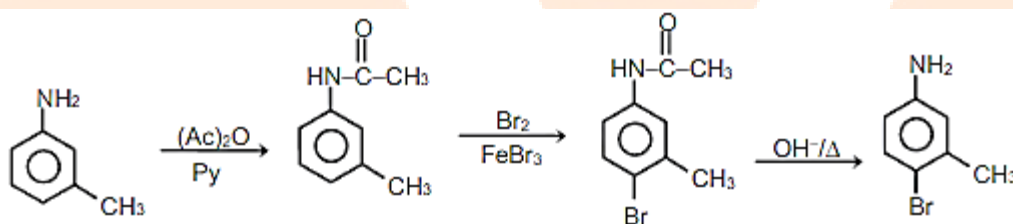


(4)

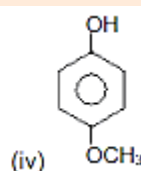
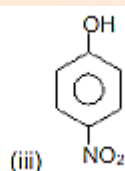
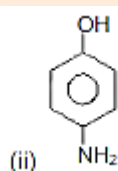
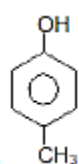


Answer: (2)

Solution:



Question 23. Which is the correct order of Boiling point of given compounds?



(1) (ii) > (iii) > (iv) > (i)

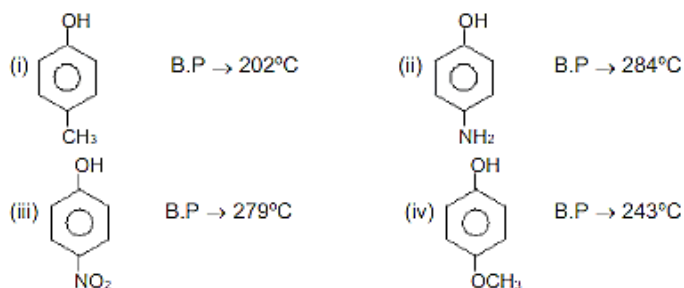
(2) (ii) > (i) > (iv) > (iii)

(3) (i) > (iii) > (iv) > (ii)

(4) (i) > (iii) > (ii) > (iv)

Answer: (1)

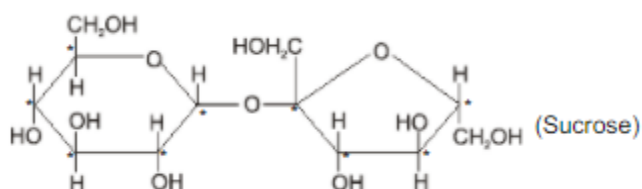
Solution:



Question 24. Total number of chiral carbon atoms present in sucrose is

Answer: 9

Solution:



Question 25. The minimum distance between the center of two octahedral voids in FCC lattice in terms of edge length is:

(1) a

(2) $\frac{a}{2}$

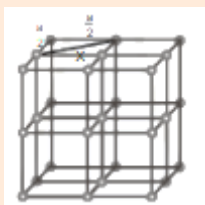
$$(3) \frac{a}{\sqrt{2}}$$

$$(4) \sqrt{2}a$$

Answer: (3)

Solution:

In FCC octahedral voids are present at the edge centers and body center.



Minimum distance between centers of two octahedral voids

$$= x = \sqrt{\left(\frac{a}{2}\right)^2 + \left(\frac{a}{2}\right)^2} = \sqrt{\frac{a^2}{4} + \frac{a^2}{4}} = \frac{a}{\sqrt{2}}$$

Question 26. Which of the following statement is correct regarding probability density (except infinity)

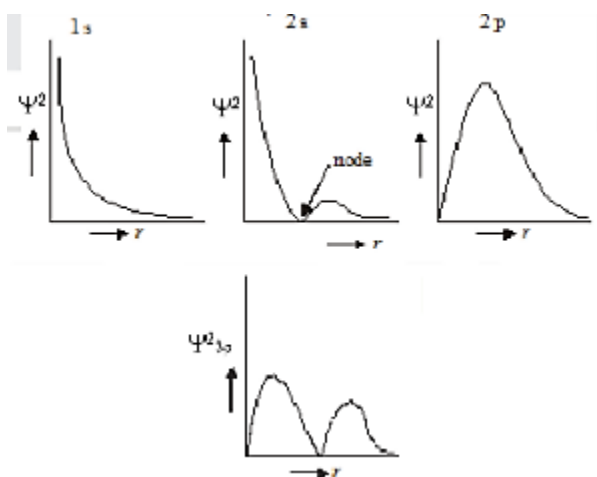
- (1) For 2p cannot be zero
- (2) For 3p can be zero
- (3) For 1s can be zero
- (4) For 2s never be zero

Answer: (2)

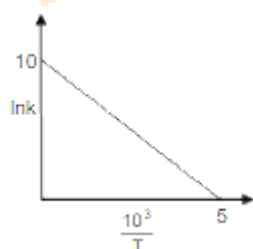
Solution:

From the following Ψ^2 function graph (Ψ^2 = probability density)

Ψ^2 can be zero for 3p orbital other than infinity.



Question 27. Using following graph find activation energy (in kJ)



- (1) R
- (2) $\frac{1}{R}$
- (3) $2R$
- (4) $\frac{2}{R}$

Answer: (3)

Solution:

$$k = Ae^{-E_a/RT}$$

$$\ln k = \ln A - \left(\frac{E_a}{R} \right) \frac{1}{T}$$

$$\ln k = \ln A - \left(\frac{E_a}{R \times 10^3} \right) \times \frac{10^3}{T}$$

Slope of graph

$$= \frac{-E_a}{R \times 10^3} = \frac{-10}{5}$$

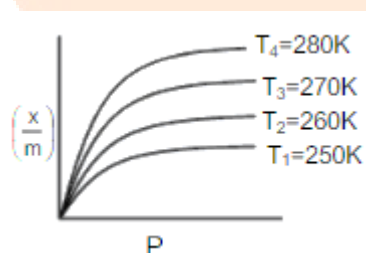
$$-E_a = 2R \times 10^3 \text{ J}$$

$$E_a = 2R \text{ kJ}$$

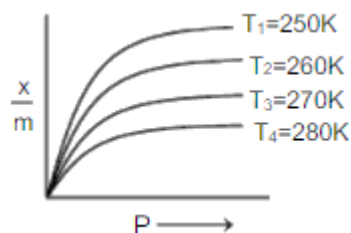
Question 28. Which of the following graph between $\left(\frac{x}{m} \right)$

Vs P is correct?

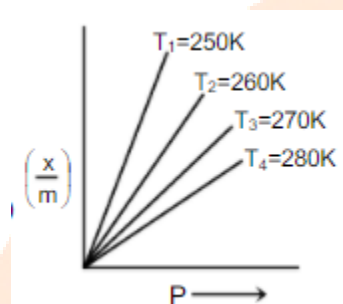
(1)



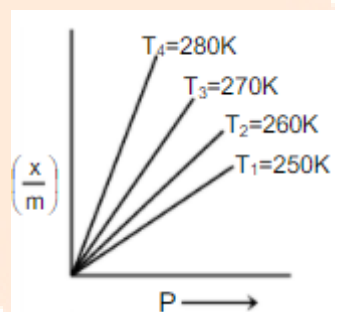
(2)



(3)



(4)



Answer: (2)

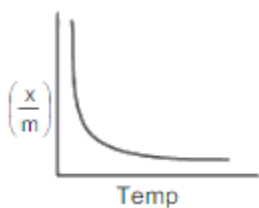
Solution:

From Freundlich adsorption isotherm

$$\frac{x}{m} \propto P \quad (\text{At low pressure})$$

$$\frac{x}{m} \propto P^0 \quad (\text{At high pressure})$$

→ On increasing temperature physical adsorption decreases



Question 29. A diatomic gas expands adiabatically so that final density is 32 times the initial density. Final pressure becomes N times of initial pressure. The value of N is

(1) 128

(2) $\frac{1}{32}$

(3) 32

(4) $\frac{1}{128}$

Answer: (1)

Solution:

1. Since the data of temperature is not given. It should be reversible adiabatic process

2. Since it is diatomic gas, we can take degree of freedom = 5

$$\gamma = \frac{f+2}{f} = \frac{7}{5}$$

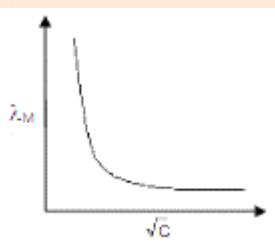
3. Use $PV^\gamma = K$

$$P_1 V_1^\gamma = P_2 V_2^\gamma \Rightarrow \frac{P_2}{P_1} = \left(\frac{V_1}{V_2} \right)^\gamma$$

$$\text{Given: } \frac{d_2}{d_1} = 32 = \frac{\frac{m}{V_2}}{\frac{m}{V_1}} = 32 \Rightarrow \frac{V_1}{V_2} = 32$$

$$\frac{P_2}{P_1} = (32)^{\frac{1}{5}} = [2^5]^{\frac{1}{5}} = 2^1 \Rightarrow \frac{P_2}{P_1} = 2 \Rightarrow P_2 = 2P_1$$

Question 30. Following graph is observed for which of the electrolytic solution



- (1) CH_3COOH
- (2) HCl
- (3) KNO_3
- (4) NaCl

Answer: (1)

Solution:

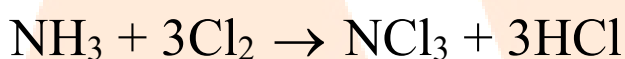
Graph represent variation of λ_M^c with respect to $\sqrt{C}$ for weak electrolyte.

Question 31. The products formed by reaction of ammonia with excess of Chlorine are:

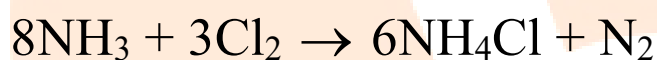


Answer: (3)

Solution:



Limiting excess



excess limiting

Question 32. (A) = $\text{cis}[\text{Co}(\text{en})_2\text{Cl}_2]^{\oplus}$ & (B) = $\text{trans}[\text{Co}(\text{en})_2\text{Cl}_2]^{\oplus}$

Which of the above complexes is optically active?

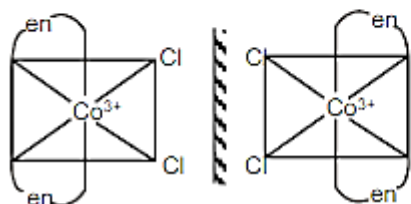
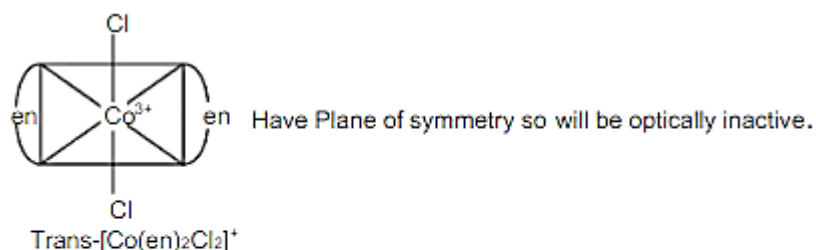
(1) Only A

(2) Only B

(3) Both

(4) None

Solution:



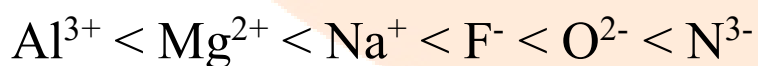
cis-[Co(en)₂Cl₂]⁺ → is optically active without plane of symmetry.

Question 33. Correct arrangement of the following species in the increasing order of their size is: N³⁻, O²⁻, F⁻, Na⁺, Mg²⁺, Al³⁺

- (1) Al³⁺, Mg²⁺, Na⁺, F⁻, O²⁻, N³⁻
- (2) N³⁻, O²⁻, F⁻, Na⁺, Mg²⁺, Al³⁺
- (3) Al³⁺, Mg²⁺, Na⁺, N³⁻, O²⁻, F⁻
- (4) Na⁺, Mg²⁺, Al³⁺, F⁻, O²⁻, N³⁻

Answer: (1)

Solution:



All are isoelectronic species so more is the z_{eff} less will be the ionic size.

Question 34. Which of the following has maximum bond angle [consider C, N, O, S as central atom]

- (1) H_2O
- (2) H_2S
- (3) NH_2
- (4) CH_4

Answer: (4)

Solution:



sp^3 , $\ell.p = 2$, B.A. = $104^\circ 30'$

$\text{H}_2\text{S} \rightarrow$ No hybridization [Drago's rule], bond angle = 90°

$\ddot{\text{N}}\text{H}_3$ sp^3 , $\ell.p = 1$, B.A. = 107°

CH_4 sp^3 , $\ell.p = 0$, B.A. = $109^\circ 28'$

Question 35. In pure form H_2O_2 is found as:

- (1) Linear, Blue colour
- (2) Linear, Colourless
- (3) Planar, Blue colour
- (4) Non planar, Blue colour

Answer: (4)

Solution:

Structure of H_2O_2 is of open book shape. It is a colour less viscous liquid but in large quantity appears blue in colour.

Question 36. Pure boron and silicon can be obtained by

- (1) Electrolytic refining
- (2) Vapour phase refining
- (3) Zone refining
- (4) Mond's process

Answer: (3)

Solution:

Germanium, Silicon, Boron, Gallium and Indium can be obtained in pure state by zone refining process.

Question 37. 0.02 M $\text{K}_2\text{Cr}_2\text{O}_7$ is treated with 0.288 gram of Ferrous oxalate. How much volume (in mL) of $\text{K}_2\text{Cr}_2\text{O}_7$ is required?

Answer: 100

Solution:

Milliequivalent of $\text{K}_2\text{Cr}_2\text{O}_7$ = Milliequivalent of FeC_2O_4

$$V_{\text{ml}} \times 0.02 \times 6 = \frac{0.288}{144} \times 3 \times 1000$$

$$V_{\text{ml}} \times 0.02 \times 6 = 2 \times 6$$

$$V_{\text{ml}} = 100 \text{ ml}$$

Question 38. For the reaction, $2A(g) \rightarrow A_2(g)$ following data is obtained at 298 K. $\Delta U = -20 \text{ kJ}$, $\Delta S = -30 \text{ J}$ then find ΔG (in kJ)

Answer: -13.5

Solution:

From $\Delta H = \Delta U + \Delta n_g RT$

$$\begin{aligned}\Delta H &= -20 \times 1000 - 1 \times 8.314 \text{ J/mol. K} \times 298 \text{ K} \\ &= -22477.572 \text{ J}\end{aligned}$$

$$\Delta G = \Delta H - T\Delta S$$

$$= -13537.572 \text{ J}$$

$$= -13.5 \text{ kJ}$$

Question 39. For the reaction, $x + y \rightleftharpoons 2z$ initially 1 mol of x, 1.5 mole of y and 0.5 mole z are taken, then at equilibrium 1 mole of z is obtained, If $k_{eq} = \frac{X}{15}$ then, find the value of 'X'.

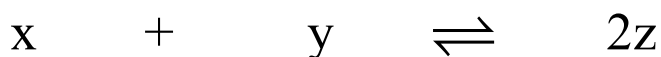
Answer: 16.00

Solution:



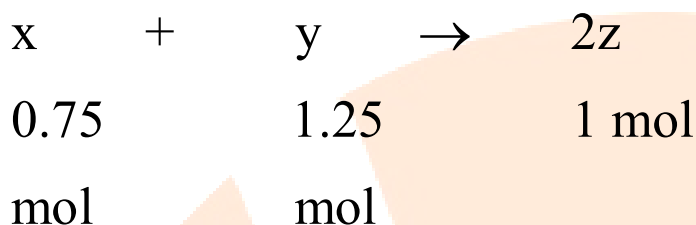
$$t = 0 \quad 1 \text{ mol} \quad 1.5 \text{ mol} \quad 0.5 \text{ mol}$$

Since moles of Z are increased at equilibrium therefore reaction goes in forward direction to attain the equilibrium.



$$t = t_{eq} \quad 1 - a \quad 1.5 - a \quad 0.5 + 2a = 1 \text{ mole}$$

$$\Rightarrow a = 0.25$$



$$K_{eq} = \frac{[z]^2}{[x][y]} = \frac{1}{0.75 \times [1.25]} = \frac{X}{15}$$

$$X = \frac{15}{(0.75 \times 1.25)} = 16$$

Mathematic

Question 40. If α, β are roots of equation $7x^2 - 3x + 2 = 0$

then find the value of $\frac{\alpha}{1-\alpha^2} + \frac{\beta}{1-\beta^2}$

(1) $\frac{7}{24}$

(2) $\frac{5}{24}$

(3) $\frac{24}{5}$

$$(4) \frac{24}{7}$$

Ans. (2)

$$\text{Sol. } \alpha + \beta = \frac{3}{7}, \alpha\beta = \frac{2}{7}$$

$$\begin{aligned} \frac{\alpha}{1-\alpha^2} + \frac{\beta}{1-\beta^2} &= \frac{(\alpha + \beta) - \alpha\beta(\alpha + \beta)}{(1-\alpha^2)(1-\beta^2)} = \frac{(\alpha + \beta) - \alpha\beta(\alpha + \beta)}{1 + (\alpha\beta)^2 - (\alpha^2 + \beta^2)} \\ &= \frac{(\alpha + \beta) - \alpha\beta(\alpha + \beta)}{1 + (\alpha\beta)^2 - (\alpha + \beta)^2 + 2\alpha\beta} = \frac{\frac{3}{7} - \frac{2}{7}\left(\frac{3}{7}\right)}{1 + \left(\frac{2}{7}\right)^2 - \left(\frac{3}{7}\right)^2 + 2\left(\frac{2}{7}\right)} \Rightarrow \frac{\left(\frac{15}{49}\right)}{\left(\frac{72}{49}\right)} = \frac{15}{72} \end{aligned}$$

Question 41. If $\vec{a}, \vec{b}, \vec{c}$ are three vectors such that

$|\vec{a}| = 2, |\vec{b}| = 4, |\vec{c}| = 4, \vec{b} \cdot \hat{a} = \vec{c} \cdot \hat{a}$ then find the value of

$$|\vec{a} + \vec{b} - \vec{c}|$$

(1) 6

(2) $\sqrt{6}$

(3) 7

(4) $2\sqrt{6}$

Ans. (1)

Sol. Given, $\vec{b} \cdot \hat{a} = \vec{c} \cdot \hat{a}$

$$\Rightarrow \frac{\vec{b} \cdot \vec{a}}{|\vec{a}|} = \frac{\vec{c} \cdot \vec{a}}{|\vec{a}|} \Rightarrow \vec{b} \cdot \vec{a} = \vec{c} \cdot \vec{a}$$

$$|\vec{a} + \vec{b} - \vec{c}|^2 = |\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 + 2(\vec{a} \cdot \vec{b} - \vec{b} \cdot \vec{c} - \vec{a} \cdot \vec{c})$$

$$= 4 + 16 + 16 + 2(\vec{a} \cdot \vec{b} - 0 - \vec{a} \cdot \vec{c}) = 36$$

$$\Rightarrow |\vec{a} + \vec{b} - \vec{c}| = 6$$

Question 42. If the line $x + 2y = 3$ cuts a chord of length r unit with the circle $x^2 + y^2 = r^2$ then find r^2 .

(1) $\frac{12}{5}$

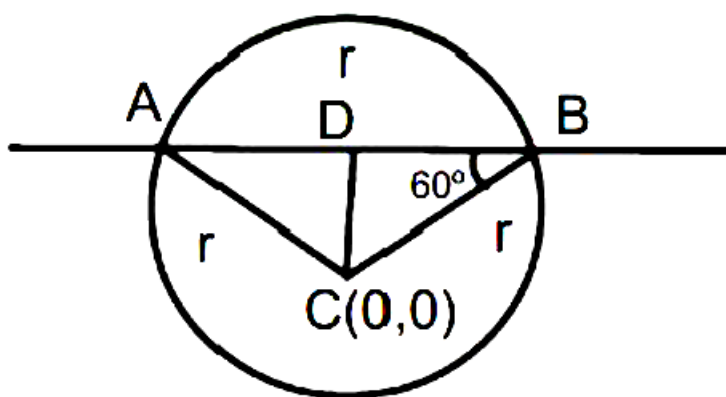
(2) $\sqrt{12}$

(3) $\frac{5}{12}$

(4) $\sqrt{\frac{12}{5}}$

Ans. (1)

Sol. $AB = r$, $AD = \frac{r}{2}$



$$CD = r \sin 60^\circ = \frac{\sqrt{3}r}{2}$$

$$\Rightarrow \frac{|0 + 0 - 3|}{\sqrt{1^2 + 2^2}} = \frac{\sqrt{3}r}{2} \Rightarrow r = 2\sqrt{\frac{3}{5}} \Rightarrow r^2 = \frac{12}{5}$$

Question 43. Find the coefficient of x^4 in the expansion of $(1 + x + x^2 + x^3)^6$

- (1) 100
- (2) 110
- (3) 120
- (4) 125

Ans. (3)

$$\text{Sol. } (1 + x + x^2 + x^3)^6 = (1 + x)^6 \cdot (1 + x^2)^6$$

$$= \left(\sum_{r=0}^6 {}^6C_r \cdot x^r \right) \cdot \left(\sum_{r=0}^6 {}^6C_r \cdot x^{2r} \right)$$

$$\text{Coefficient of } x^4 = {}^6C_0 \cdot {}^6C_2 + {}^6C_2 \cdot {}^6C_1 + {}^6C_4 \cdot {}^6C_0$$

$$\text{Coefficient of } x^4 = 15 + 90 + 15 = 120$$

Question 44. If the mean and standard deviation of 5, 3, 7, a, b are 5 and 2 respectively, then a and b are roots of equation.

(1) $x^2 - 10x + 18 = 0$

(2) $x^2 - 20x + 18 = 0$

(3) $x^2 - 20x + 19 = 0$

(4) $x^2 - 10x + 19 = 0$

Ans. (4)

Sol. $\bar{x} = \frac{5 + 3 + 7 + a + b}{5}$

$5 + 3 + 7 + a + b = 25 \Rightarrow a + b = 10$

S.D. = $\sqrt{\frac{5^2 + 3^2 + 7^2 + a^2 + b^2}{5}} - 5^2 = 2$

$\Rightarrow \frac{a^2 + b^2 + 83}{5} - 25 = 4 \Rightarrow a^2 + b^2 = 62$

$\Rightarrow (a + b)^2 - 2ab = 62 \Rightarrow ab = 19$

So, equation whose roots are a and b is $x^2 - 10x + 19 = 0$

Question 45. There are three section A, B, C in a paper each section having 5 questions. In how many ways a student can solve exactly 5 questions taken at least one question from each section.

(1) 2200

(2) 2225

(3) 2250

(4) 2275

Ans. (3)

Sol. $A \rightarrow 5Q$ $B \rightarrow 5Q$ $C \rightarrow 5Q$

A_1, A_2, A_3, A_4, A_5 B_1, B_2, B_3, B_4, B_5 C_1, C_2, C_3, C_4, C_5

No. of ways when 3 questions are selected from one section and one from each from other sections = ${}^3C_1 \times {}^5C_3 \times {}^5C_1 \times {}^5C_1 = 750$

No. of ways when 2 questions are selected from two sections and one from other section = ${}^3C_2 \times {}^5C_2 \times {}^5C_2 \times {}^5C_1 = 1500$

$\therefore$ total = 2250

Question 46. $\left(\frac{-1 + \sqrt{3}i}{1 - i} \right)^{30}$ simplifies to

(1) -215i

(2) 215i

(3) 215

(4) -215

Ans. (1)

$$\begin{aligned}\text{Sol. } \left(\frac{-1 + \sqrt{3}i}{1 - i} \right)^{30} &= \left(\frac{2 \cos \left(\frac{2\pi}{3} \right) + i \sin \left(\frac{2\pi}{3} \right)}{\sqrt{2} \left(\cos \frac{\pi}{4} - i \sin \frac{\pi}{4} \right)} \right) \\ &= \frac{2^{30} (\cos 20\pi + i \sin 20\pi)}{2^{15} \left(\cos \frac{15\pi}{2} - i \sin \frac{15\pi}{2} \right)} \\ &= \frac{2^{15} (1 + 0i)}{(0 + i)} = -2^{15} i\end{aligned}$$

Question 47. If the lines $x - y = a$ and $x + y = b$ are tangents for $y = x^2 - 3x + 2$ then $\frac{a}{b} =$

Ans. (2)

$$\text{Sol. } y = x^2 - 3x + 2, \quad x + y = a, \quad x - y = b$$

$$2x_1 - 3 = 1 \qquad 2x_2 - 3 = -1$$

$$x_1 = 2 \qquad x_2 = 1$$

$$y_1 = 4 - 6 + 2 = 0 \qquad y_2 = 0$$

$$(2, 0) \qquad (1, 0)$$

$$a = 2 \qquad b = 1$$

$$\therefore \frac{a}{b} = \frac{2}{1} = 2$$

Question 48. Let

$$y_1 = \tan^{-1} \left(\frac{\sqrt{1+x^2}-1}{x} \right) \text{ and } y_2 = \tan^{-1} \left(\frac{2x\sqrt{1-x^2}}{1-2x^2} \right) \text{ then } \frac{dy_1}{dy_2} =$$

(1) $\frac{\sqrt{1-x^2}}{2(1+x^2)}$

(2) $\frac{\sqrt{1-x^2}}{4(1+x^2)}$

(3) $\frac{1}{(1+x^2)\sqrt{1-x^2}}$

(4) $\frac{1}{4(1+x^2)\sqrt{1-x^2}}$

Ans. (2)

Sol. Let $x = \tan \theta$

$$y_1 = \tan^{-1} \left(\frac{\sec \theta - 1}{\tan \theta} \right) = \tan^{-1} \left(\tan \frac{\theta}{2} \right) = \frac{\theta}{2} = \frac{1}{2} \tan^{-1} x$$

Let, $x = \sin \phi$

$$y_2 = \tan^{-1} \left(\frac{2 \sin \phi \cos \phi}{\cos 2\phi} \right) = \tan^{-1} (\tan 2\phi) = 2\phi = 2 \sin^{-1} x$$

$$\frac{dy_1}{dy_2} = \frac{dy_1 / dx}{dy_2 / dx} = \frac{\frac{1}{(1+x^2)} \cdot \frac{1}{2}}{2 \cdot \frac{1}{\sqrt{1-x^2}}} \\ = \frac{\sqrt{1-x^2}}{4(1+x^2)}$$

Question 49. In a G.P sum of 2nd, 3rd and 4th term is 3 and that of 6th, 7th and 8th term is 243 then $S_{50} =$

(1) $\frac{3^{50} + 1}{26}$

(2) $\frac{3^{50} - 1}{13}$

(3) $\frac{3^{50} - 1}{26}$

(4) $\frac{3^{49} - 1}{26}$

Ans. (3)

Sol. Let a, ar, ar², G.P.

$$T_2 + T_3 + T_4 = 3 \Rightarrow ar(1 + r + r^2) = 3 \quad \dots (i)$$

$$T_6 + T_7 + T_8 = 243 \Rightarrow ar^5(1 + r + r^2) = 243 \\ \dots (ii)$$

By (i) and (ii)

$$r^4 = 81 \quad \Rightarrow \quad r = 3$$

$$\therefore a = \frac{1}{13}$$

$$S_{50} = \frac{a(r^{50} - 1)}{r - 1} = \frac{3^{50} - 1}{26}$$

Question 50. $\int \frac{\cos \theta}{7 + \sin \theta - 2 \cos^2 \theta} d\theta$ is equal to

(1) $\frac{2}{\sqrt{39}} \tan^{-1} \left(\frac{2 \sin \theta + 1}{\sqrt{39}} \right) + C$

(2) $\frac{2}{\sqrt{39}} \tan^{-1} \left(\frac{4 \sin \theta + 1}{\sqrt{39}} \right) + C$

(3) $\frac{4}{\sqrt{39}} \tan^{-1} \left(\frac{4 \sin \theta + 1}{\sqrt{39}} \right) + C$

(4) $\frac{4}{\sqrt{39}} \tan^{-1} \left(\frac{2 \sin \theta + 1}{\sqrt{39}} \right) + C$

Ans. (2)

Sol. $I = \int \frac{\cos \theta}{7 + \sin \theta - 2(1 - \sin^2 \theta)} d\theta$

$$= \int \frac{\cos \theta}{2 \sin^2 \theta + \sin \theta + 5} d\theta$$

$$\sin \theta = t \Rightarrow \cos \theta \cdot d\theta = dt$$

$$I = \int \frac{dt}{2t^2 + t + 5} = \frac{1}{2} \int \frac{dt}{\left(t + \frac{1}{4}\right)^2 + \frac{39}{16}}$$

$$\Rightarrow I = \frac{1}{2} \cdot \frac{1}{\frac{\sqrt{39}}{4}} \tan^{-1} \left(\frac{t + \frac{1}{4}}{\frac{\sqrt{39}}{4}} \right) + C$$

$$= \frac{2}{\sqrt{39}} \tan^{-1} \left(\frac{4 \sin \theta + 1}{\sqrt{39}} \right) + C$$

Question 51. The area enclosed by $[x]$. $(x-1) \leq y \leq 2\sqrt{x}$ for $x = 0$ to 2 where $[x]$ is greatest integer less than or equal to x , is equal to

(1) $\frac{8\sqrt{2}}{3} + \frac{1}{2}$

(2) $\frac{8\sqrt{2}}{3}$

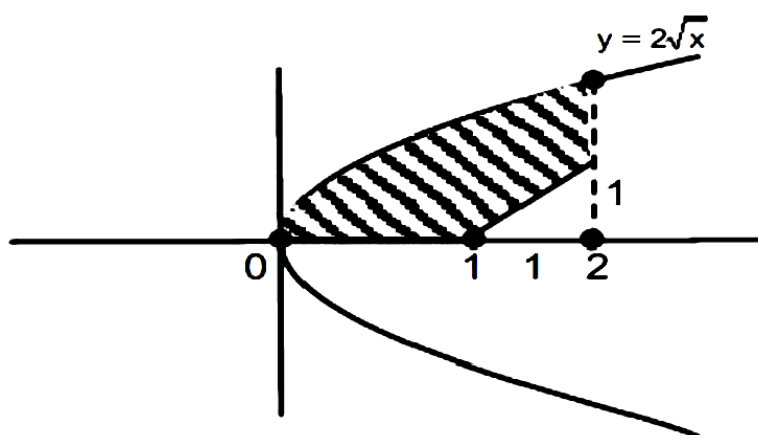
(3) $\frac{8\sqrt{2}}{3} - \frac{1}{2}$

(4) $\frac{8}{3} - \frac{1}{\sqrt{2}}$

Ans. (3)

Sol. $y = [x] (x - 1)$

$$= \begin{cases} 0 & 0 \leq x < 1 \\ x - 1 & 1 \leq x < 2 \end{cases}$$



$$\text{Required Area} = \int_0^2 2\sqrt{x} \cdot dx - \frac{1}{2}(1)(1) = \left(\frac{4x^{3/2}}{3} \right)_0^2 - \frac{1}{2} = \frac{8\sqrt{2}}{3} - \frac{1}{2}$$

Question 52. If $x + a = y + b + 1 = z + c$ then the value of

$$\begin{vmatrix} x & a+y & a+x \\ y & b+y & b+y \\ z & c+y & c+z \end{vmatrix} \text{ is}$$

- (1) $y(a - b)$
- (2) $y(b - c)$
- (3) $y(c - a)$
- (4) 0

Ans. (3)

Sol. Given $x + a = y + b + 1 = z + c$

$$\text{Now } \begin{vmatrix} x & a+y & a+x \\ y & b+y & b+y \\ z & c+y & c+z \end{vmatrix} = \begin{vmatrix} x & a+y & a \\ y & b+y & b \\ z & c+y & c \end{vmatrix} (C_3 \rightarrow C_3 - C_1)$$

$$= \begin{vmatrix} x & y & a \\ y & y & b \\ z & y & c \end{vmatrix} (C_2 \rightarrow C_2 - C_3)$$

$$= y \begin{vmatrix} x & 1 & a \\ y & 1 & b \\ z & 1 & c \end{vmatrix}$$

$$R_2 \rightarrow R_2 - R_1 \text{ and } R_3 \rightarrow R_3 - R_1$$

$$y. \begin{vmatrix} x & 1 & a \\ y-x & 0 & b-a \\ z-x & 0 & c-a \end{vmatrix}$$

$$= -y(c-a) \begin{vmatrix} x & 1 & a \\ y-x & 0 & b-a \\ -1 & 0 & 1 \end{vmatrix}$$

$$= -y.(c-a)(y-x+b-a) = y(c-a)$$

Question 53. If $\log_{\frac{1}{7^2}} x + \log_{\frac{1}{7^3}} x + \log_{\frac{1}{7^4}} x + \dots \dots \dots 20 \text{ times}$
 $= 460$ then $x = ?$

Ans. (49)

Sol. Given $\log_{7^{\frac{1}{2}}} x + \log_{7^{\frac{1}{3}}} x + \log_{7^{\frac{1}{4}}} x + \dots 20 \text{ times} = 460$

$$\Rightarrow (2 + 3 + 4 + \dots + 21) \log_7 x = 460$$

$$\Rightarrow \frac{20}{2} (2 + 21) \log_7 x = 460$$

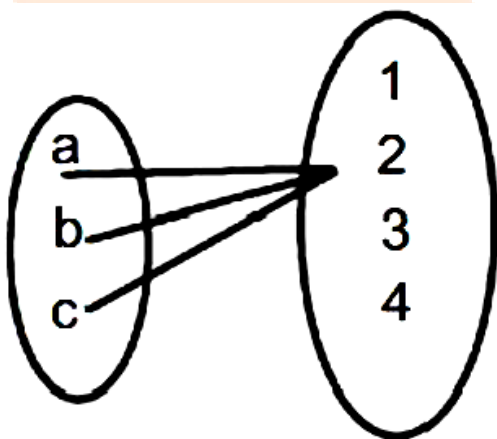
$$\Rightarrow \log_7 x = 2$$

$$\Rightarrow x = 49$$

Question 54. A function $f: A \rightarrow B$ where $A = \{a, b, c\}$, $B = \{1, 2, 3, 4\}$. How many function can be defined from A to B which are not one-one such that $2 \in f(A)$

Ans. (19)

Sol. only '2' is range $\rightarrow$ 1 function



one element out of 1, 3, 4, is in range with '2'

$$\text{Number of ways} = {}^3C_1 \cdot \frac{3!}{2! \cdot 1!} \cdot 2! = 18$$

(Select one from 1, 3, 4 and distribute among a, b, c)

$$\text{Total function} = 1 + 18 = 19$$

Question 55. If the system of equations $x + y + z = 0$, $x + 3y + k^2z = 0$ and $x + 2y + z = 0$ have a non-zero solution then

the value of $y + \frac{x}{z}$ is

(1) 1

(2) 0

(3) -1

(4) 2

Ans. (3)

Sol.

$$\left. \begin{array}{l} x + y + z = 0 \\ x + 3y + k^2z = 0 \\ x + 2y + z = 0 \end{array} \right\} \text{ has a non-zero solution.}$$

$$\therefore \begin{vmatrix} 1 & 1 & 1 \\ 1 & 3 & k^2 \\ 1 & 2 & 1 \end{vmatrix} = 0$$

$$\Rightarrow 3 - 2k^2 - 1 + k^2 + 2 - 3 \Rightarrow k^2 = 1$$

$\therefore$ system is

$$x + y + z = 0$$

$$x + 2y + z = 0$$

$$x + 3y + z = 0$$

$$\therefore y = 0, x + z = 0 \Rightarrow \frac{x}{z} = -1$$

$$\therefore y + \frac{x}{z} = -1$$

Question 56. If $y = mx + C$ is a common tangent of circle $x^2 + y^2 = 3$ and hyperbola $\frac{x^2}{64} - \frac{y^2}{100} = 1$ then which of the following statement is true:

- (1) $8m = 4$
- (2) $61C^2 = 492$
- (3) $4C^2 = 369$
- (4) $8m + 5 = 0$

Ans. (2)

Sol. Tangent of circle $\Rightarrow C^2 = 3(1 + m^2)$

and tangent of hyperbola $\Rightarrow C^2 = 64m^2 - 100$

$$\Rightarrow 3(1 + m^2) = 64m^2 - 100$$

$$\Rightarrow 61m^2 = 103$$

$$\Rightarrow m^2 = \frac{103}{61}$$

$$\Rightarrow C^2 = 3\left(1 + \frac{103}{61}\right) = \frac{492}{61}$$

